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Hierarchical frequency control strategy of doubly-fed induction diesel generator under emergency condition

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Abstract

Natural disasters lead to the uncertain topological structure of the urban distribution network. It is difficult for the communication system to recover quickly, which brings great difficulties to restoring the power supply. Therefore, the deployment of an emergency power supply is one of the critical measures to ensure the stable operation of isolated network systems and reduce power loss. Compared with the traditional synchronous diesel generator (SDG) as an emergency power supply, a doubly-fed induction diesel generator (DFIDG) has the advantages of flexible and controllable, rapid response and high operation efficiency. This paper presents a layered frequency control strategy for a doubly-fed induction diesel generator in an emergency scenario. Firstly, the models of doubly-fed induction diesel generator and its control system, prime mover and its speed regulation system are established. Secondly, the hierarchical frequency control strategy is proposed, including primary integrated frequency control (including droop control and virtual inertia control) and secondary frequency regulation control based on translation method. The virtual inertia control can make the doubly-fed machine quickly release the rotor kinetic energy when the frequency changes rapidly, so as to improve the frequency supportability. According to the overall shortage of the system and the maximum output power of the doubly-fed induction diesel generator, the strategy calculates the secondary frequency regulation power adjustment as the DFIDG power setting value, and then can realize automatic secondary frequency regulation without a communication system. Finally, simulation results verify the effectiveness of the proposed strategy.
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Keywords: Natural disaster; Variable speed doubly-fed induction diesel generator; Hierarchical frequency control; Virtual inertia control

1. Introduction

The low-voltage distribution network is prone to significant power outages after natural disasters (typhoons, ice and snow, earthquakes, etc.). In order to avoid excessive power outages for essential users and reduce economic

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losses, urban distribution networks are often equipped with mobile emergency power sources with a specific capacity (such as static energy storage, diesel generators (DG), etc.). Due to the mature technology and low price, the traditional synchronous diesel generators (SDG) are used as the emergency power supply in most cases. Still, its disadvantages are obvious: first, after losing the support of the bulk power system, the stability of the isolated network formed by the traditional SDG and load is very poor. Secondly, when the SDG operates under light load, insufficient combustion leads to low operation efficiency, poor economy and environmental pollution. Anthony L. Rogers [1] proposed the concept of a variable speed DG in 1996, proving that the variable speed DG is more efficient than the fixed speed SDG. In the isolated network system, when the DG runs at a low load, the fuel consumption of the variable-speed unit is reduced by more than 40% compared with that of the fixed-speed unit. In addition, the variable speed unit also has the advantages of good regulation characteristics and high stability. Compared with direct-driven variable speed units, doubly-fed variable speed units have been widely used in wind power, hydropower and other fields because of their advantages of small power converter capacity [2,3]. At present, the research of variable speed doubly-fed induction diesel generators has gradually become a hot spot [4].

At present, the research on DG in emergency power supply mainly focuses on the capacity configuration and dispatch of generator vehicles, the location of access points, and the coordinated control of power. For example, literature [5] proposes an optimal allocation model of the emergency power supply according to the geographical division of the power supply substation and the importance of users. Literature [6] proposed a two-stage mobile emergency power generation vehicle and fuel supply model, which transformed the problem into a hybrid generator assignment and route optimization model, linearized by large M method and variable replacement, then converted into a mixed-integer linear programming problem. Literature [7] proposed an optimal island search algorithm based on graph theory to reduce the economic loss and established the optimal search of the mobile emergency power access point and the multi constraint optimization model of the power access point in the island. Literature [8] determines the inverter to work at maximum power or limited power based on the grid-side frequency detected by the phase-locked loop. It proposes a traditional SDG no-communication power adaptive control strategy based on active frequency droop control.

In summary, there is no specific research on power and frequency coordinated control of DFIDG under emergency power supply applications. In this paper, the doubly-fed generator model, diesel engine governor and optimal speed model are established, respectively. The layered power control strategy of the doubly-fed variable speed diesel engine in an emergency environment is proposed, including the primary integrated frequency control composed of droop control and virtual inertia control and the secondary frequency control of the outer layer. Finally, a simulation model is established based on MATLAB/Simulink to verify the effectiveness of the proposed control strategy. Finally, a simulation model is established based on Matlab/Simulink to verify the effectiveness of the proposed control strategy.

2. Modelling of DFIDG

DFIDG is a new type of diesel generator and mainly composed of diesel engine, digital governor, mechanical-hydraulic servo actuator, optimal speed module, AC-excitation motor, back-to-back converter, converter controller, frequency regulator, etc. It can be divided into two parts: the diesel engine system and the double-fed generator system. The model block diagram of the system is shown in Fig. 1.

RSC is the rotor side converter. GSC is the grid side converter. Q_g is the reactive power of the DC bus of the back-to-back converter. U_{dc} is the voltage of the DC bus. T_m is the mechanical torque of the diesel engine. T_e and P_e are the electromagnetic torque and electromagnetic power of the doubly-fed generator. n_r, n_{ref} and Δn are actual speed, reference speed and speed difference of diesel engine rotor. P, P_{ref}, Q and Q_{ref} are the actual and reference values of active and reactive power, respectively, ΔP is the additional reference power, P_{set} is the set power, f, Δf and f_{ref} are actual frequency, frequency difference and reference frequency, respectively.

2.1. Modelling of the doubly-fed induction generator system

The stator-side of the doubly-fed machine is connected to the power grid or load, and the grid-side converter is responsible for maintaining the voltage stability at the DC bus side. The rotor adopts three-phase symmetrically distributed excitation winding, and the rotor-side converter changes the amplitude and phase of the excitation voltage according to the power command. The active and reactive power decoupling control of a doubly-fed machine is

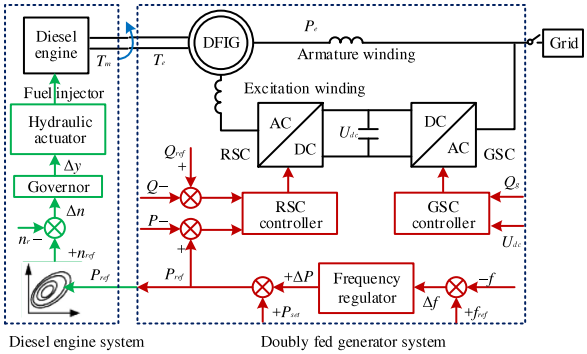


Fig. 1. Structure diagram of variable speed doubly-fed induction diesel generator system.

realized by controlling the size of the excitation magnetic field. The motor speed can be controlled by changing the excitation current frequency and the slip. The doubly-fed machine model and the rotor converter control model is established, respectively.

2.1.1. modelling of doubly-fed induction generator

The doubly-fed machine with the electromagnetic transient process can be represented by the full state 5th-order model [9], and only the algebraic equations related to control are given here. In d–q coordinates, when the stator flux-oriented vector control strategy is adopted for the doubly-fed machine, the equations of the active power, reactive power and electromagnetic torque expressed by rotor current are as follows:

$$\begin{cases} P_s = \dfrac{3}{2}\left(u_{ds}i_{ds} + u_{qs}i_{qs}\right) = \dfrac{3}{2}U_s i_{qs} = \dfrac{3L_m}{2L_s}U_s i_{qr} \\ Q_s = \dfrac{3}{2}\left(u_{qs}i_{ds} - u_{ds}i_{qs}\right) = \dfrac{3}{2}U_s i_{ds} = \dfrac{3}{2}U_s\dfrac{L_m i_{dr} - \psi_s}{L_s} = \dfrac{3L_m}{2L_s}U_s i_{dr} - \dfrac{3U_s^2}{2\omega L_s} \\ T_e = n_p(\psi_{ds}i_{qs} - \psi_{qs}i_{ds}) = n_p\psi_{ds}i_{qs} = n_p\psi_s \times \dfrac{L_m}{L_s} \times i_{qr} = \dfrac{n_p L_m}{\omega L_s}U_s i_{qr} \end{cases} \tag{1}$$

where, P_s, Q_s, T_e, u_{ds}, u_{qs}, i_{ds}, i_{qs}, i_{dr}, i_{qr}, U_s, ψ_s, ω, L_m and L_s are generator stator active power, stator reactive power, electromagnetic torque, stator voltage direct axis component, stator voltage quadrature axis component, stator current direct axis component, stator voltage quadrature axis component, rotor current quadrature axis component, stator voltage synthesis vector, stator flux synthesis vector, d–q coordinate rotation angular velocity self-inductance coefficient of stator equivalent two-phase winding in d–q coordinate and mutual inductance coefficient between stator and rotor coaxial winding in d–q coordinate.

2.1.2. Control model of rotor-side converter

Under the stator flux orientation, the active and reactive power of the doubly-fed machine is adjusted by the stator current d-axis and q-axis components i_{ds} and i_{qs} respectively. The system adopts a double closed-loop control structure. The outer loop is the power control loop, and the inner loop is the current control loop. The current inner loop control generates the modulation wave by generating the three-phase voltage component of the rotor and is used for SVPWM trigger control. The rotor side control of the doubly-fed converter using stator flux-oriented vector control is shown in Fig. 2.

P_{ref} is the reference value of active power at the machine-side. The additional power can be considered in the active power setting value, so as to realize the droop control, virtual inertia control and secondary frequency regulation.

2.2. Modelling of diesel engine system

2.2.1. Diesel engine model

The diesel engine converts chemical energy into mechanical energy. Its rotating shaft is connected with the doubly-fed machine to drive the rotor of the induction generator to generate electricity. Since fuel combustion is

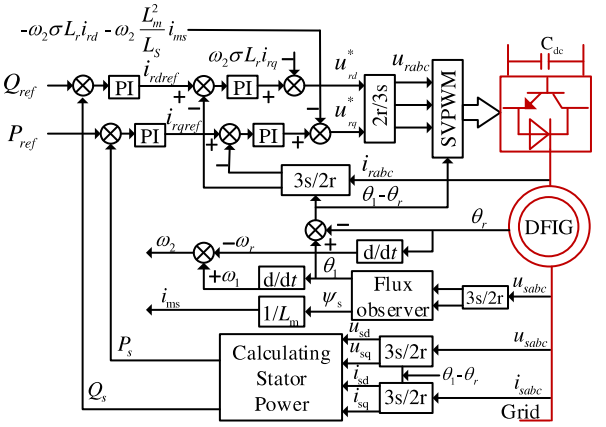


Fig. 2. Diagram of rotor side control for DFIG converter.

a multi-physical process and a high-dimensional nonlinear complex model, when more attention is paid to the electromechanical characteristics of diesel engines, its thermodynamic process can be described by a first-order function including engine gain and combustion delay. The system structure block diagram is shown in Fig. 3, mainly composed of engine, governor, hydraulic actuator, etc. The governor dynamically tracks the optimal speed according to the speed deviation. The hydraulic actuator amplifies the reference electrical signal and controls the fuel valve for fuel injection. The amount of fuel injection determines the mechanical power of the unit. A simplified mathematical model of the diesel engine, as shown in Eq. (2), is established.

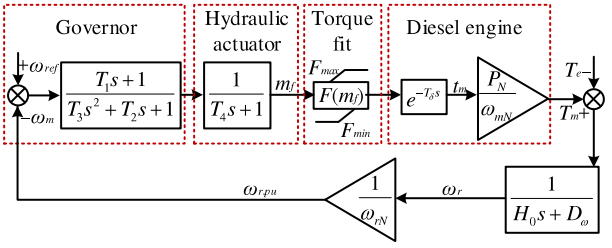


Fig. 3. System structure block diagram.

$$\begin{cases} t_{\mathrm{m}} = f\left(m_{\mathrm{f}}\right) \times e^{-T_{\delta} \times s} / \left(1 + T_{\mathrm{ha}} s\right) \\ F\left(m_{\mathrm{f}}\right) = d_3 \times m_{\mathrm{f}}^3 + d_2 \times m_{\mathrm{f}}^2 + d_1 \times m_{\mathrm{f}} + d_0 \\ T_{\delta} = 60 \times S_{\mathrm{t}} / \left(2 \times N \times n_{\mathrm{cy}}\right) + 60/(4 \times N) \end{cases} \tag{2}$$

where, t_{m} is the unit value of mechanical torque of diesel engine, T_{ha} is the time constant of a hydraulic actuator, T_{δ} is the combustion delay of the engine mechanical torque, and m_{f} is the unit value of the fuel injected by the injection valve, $F(m_{\mathrm{f}})$ is a polynomial torque-fit equation representing a variant engine torque gain. $F_{\min}$ and $F_{\max}$ are the upper and lower limits of $F(m_{\mathrm{f}})$. According to literature [10], their fitting coefficients are: d_3 = -0.109, $d_2 = 0.073$, $d_1 = 1.17$, d_0=−0.13. S_{t} is 4, indicating the four-stroke engine, n_{cy} indicates the number of cylinders, N is the rated speed of the engine, and T_1, T_2 and T_3 are the control time constants of the main controller of the governor [11].

In addition, the rotor motion equation can be expressed as:

$$2H_0 \frac{d\omega_r}{dt} = T_{\mathrm{m}} - T_{\mathrm{e}} - D_{\omega}\omega_r \tag{3}$$

where, H_0 and D_{ω} is the inertia time constant and damping constant of the DFIDG. ω_{r}, ω_{ref}, ω_{rN} and $\omega_{\mathrm{r.pu}}$ is the actual value, reference value, rated value and the unit value of the actual speed of the rotor respectively.

2.2.2. Optimal speed model

According to the load characteristic curves at different speeds, the relationship between fuel consumption and load torque and power is obtained, that is, the universal characteristic curve of diesel engines [12], as shown in Fig. 4.

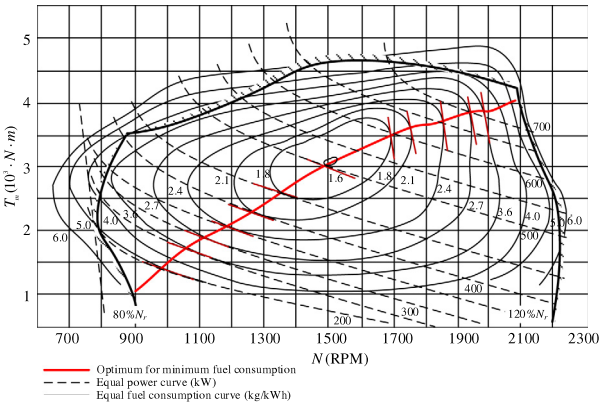


Fig. 4. Universal characteristic curve of the diesel engine.

Based on the universal characteristic curve, it can be seen that there is a set of optimal combinations of speed and power. A specific load (black dotted line in Fig. 4) will correspond to an optimal speed, and the fuel consumption is the minimum at this time. Connect all one-to-one corresponding points to get an optimal speed–power curve (red curve in Fig. 4). During the actual operation of the unit, the optimal speed under the reference power is taken as the speed reference value of the governor. By adjusting the fuel injection valve, the rotor mechanical speed can track the optimal speed in real-time to ensure the maximum combustion efficiency of the diesel generator.

3. Hierarchical frequency control strategy

The power outage area of the urban distribution network caused by disasters is uncertain, and the communication system in the power outage area needs to be established temporarily. This process is time-consuming, and reliability is difficult to guarantee. Therefore, the control strategy of emergency microgrid should have the following characteristics: (1) relying as little as possible on communication system; (2) the system load can be connected and removed stably; (3) the system frequency can be maintained within the supply capacity range. Therefore, the proposed hierarchical frequency control strategy of DFIDG under emergency conditions is shown in Fig. 5, including primary integrated frequency control (based on droop control and virtual inertia control) and secondary frequency control.

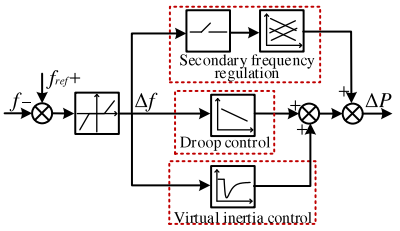


Fig. 5. Hierarchical frequency control strategy.

3.1. Primary integrated frequency control

The governor of the traditional synchronous unit adopts the droop control method to maintain frequency stability by automatically adjusting output. Similarly, adding a droop control loop to the rotor side control outer loop of the

doubly-fed converter can make the unit show droop characteristics and automatically participate in the primary frequency regulation. The droop characteristics of DFIDG are shown in Fig. 6. When the active load increases and the frequency decreases, the unit output increases correspondingly to restrain the frequency drop and vice versa. The power–frequency characteristics of droop control for a single unit and M units in an isolated grid system are as follows:

$$P_{\mathrm{G}} = P_{set} + K_{\mathrm{G}}(f_n - f) \tag{4}$$

$$P_{\Sigma G} = \sum_{i=1}^{m} (P_{seti} + (f_n - f)\, K_{\mathrm{Gi}}) \tag{5}$$

where, K_{G} is the droop coefficient, $K_{\mathrm{G}} = (P\%P_{\mathrm{N}})/(f\%f_{\mathrm{N}})$. P_{N} is the rated power of the unit. That is, the frequency decreases by $f\%$ and the active power increases by $P\%$. P_{set} is the set value of DFIDG output power when the system frequency is f_{N}, and m is the number of DFIDG in the emergency isolated network system. The $P\%/f\%$ of the same type of DFIDG is generally the same. Therefore, when the system has frequency fluctuation, the output power of DFIDG will apportion the load power according to the proportion of rated power. Namely

$$\Delta P_{\mathrm{G}} = K_{\mathrm{G}} \Delta f \tag{6}$$

$$\Delta P_{\mathrm{G1}} : \Delta P_{\mathrm{G2}} : \cdots : \Delta P_{\mathrm{Gm}} = K_{\mathrm{G1}} : K_{\mathrm{G2}} : \cdots : K_{\mathrm{Gm}} = P_{\mathrm{N1}} : P_{\mathrm{N2}} : \cdots : P_{\mathrm{Nm}} \tag{7}$$

Adding the power related to the derivative of frequency change, such as Eq. (8), release the kinetic energy stored in the rotor to actively respond to the system frequency change, which can simulate the characteristics of the conventional generator to increase the system inertia.

$$\Delta P_{\mathrm{V}} = K_{\mathrm{V}} \frac{\mathrm{d}\Delta f}{\mathrm{d}t} \tag{8}$$

where, ΔP_{V} is the power released by the rotor in response to frequency change under virtual inertia control, and K_{V} is the differential coefficient, that is, the virtual inertia coefficient. The typical frequency variation curve is shown in Fig. 7 [13].

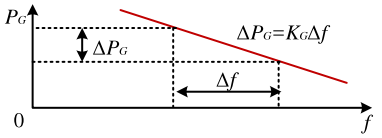


Fig. 6. Characteristic of droop control.

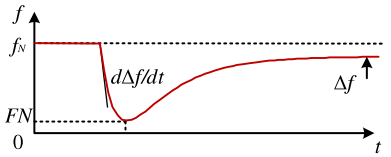


Fig. 7. Typical frequency variation curve.

When the load increases suddenly, the frequency drops rapidly and reaches the frequency nadir (FN), $\mathrm{d}\Delta f/\mathrm{d}t$ increases from 0 to the maximum, then falls back in a short time and decreases to 0 at FN. In this stage, $\mathrm{d}\Delta f/\mathrm{d}t < 0$. At this time, it is necessary to increase the inertia by increasing the power reference value of the rotor converter so that the rotor can quickly release kinetic energy, suppress the falling speed of the system frequency and improve FN.

Additional active power of primary integrated frequency control ΔP can be decomposed into proportional and differential quantities related to frequency, corresponding to droop frequency regulation characteristics and inertia response characteristics of traditional synchronous motor, respectively, which can be expressed as

$$\Delta P = \Delta P_{\mathrm{G}} + \Delta P_{\mathrm{V}} = K_{\mathrm{G}} \Delta f + K_{\mathrm{V}} \frac{\mathrm{d}\Delta f}{\mathrm{d}t} \tag{9}$$

It can be seen from Eq. (7) that when multiple DFIDGs adopt droop control, the steady-state power imbalance can be alleviated, and the frequency change can be suppressed. As seen from Eq. (8), after introducing the virtual inertia control with the differential link, the unit can quickly release the rotor kinetic energy in case of frequency disturbance, which plays an inertial support role for the system. After the unit completes the inertial response, the unit no longer provides power support. It does not continuously share the unbalanced power of the system, which has little impact on the static frequency deviation of the system. Therefore, the power distribution proportion under the transient state is affected by the differential link coefficient. However, the power variation of DFIDGs and other units under steady state are still distributed according to the droop coefficient [13].

3.2. Primary integrated frequency control

When the total power shortage of the system is significant, the secondary frequency regulation function needs to be enabled. There are three main methods of secondary frequency regulation: translation method, changing droop coefficient method and adjusting power coefficient method. Literature [14] proves that the translation method has the best adjustment characteristics and the highest sensitivity through theoretical analysis. The translation method is selected in this paper. As shown in Fig. 8, the initial stability of the system is at point A, that is, the intersection of load characteristic curve P_{LD1} and generator unit comprehensive power frequency characteristic curve P_{G1}. When the system load suddenly increases, the load characteristic curve moves from P_{LD1} to P_{LD2}, and its intersection with P_{G1} is B_1. The generator's output rises from P_{setA} to P_{GB}, and the system frequency also decreases to f_{B}. This process is primary frequency regulation. At this time, the power balance expression is Eq. (10). When f_{B} is less than the minimum frequency $F_{\min}$ of the system, secondary frequency regulation is required to make the frequency return to the normal range. Change the P_{set} of the DFIDG and shift P_{G1} upward to P_{G2}. The intersection of P_{G2} and P_{LD2} is C_1, which is a new balance point (frequency is f_{C1}). The power balance relationship is as shown in Eq. (11). The setting value of P^{C}_{seti} of each DFIDG is shown in Eq. (12).

$$P^{B}_{L} = P^{B}_{\Sigma G} = \sum_{i=1}^{m} \left(P^{A}_{\mathrm{seti}} + (f_{\mathrm{n}} - f_{\mathrm{C}})\, K_{G\mathrm{i}} \right) \tag{10}$$

$$P^{C}_{L} = P^{C}_{\Sigma G} = \sum_{i=1}^{m} \left(P^{C}_{\mathrm{seti}} + (f_{\mathrm{n}} - f_{\mathrm{C}})\, K_{G\mathrm{i}} \right) \tag{11}$$

$$P^{C}_{\mathrm{seti}} = \left(P_{\mathrm{Ni}} / \sum_{i=1}^{m} P_{\mathrm{Ni}} \right) \sum_{i=1}^{m} \left(P^{C}_{\mathrm{seti}} \right) \tag{12}$$

The maximum power P_{Gmaxi} of DFIDG determines the value range of f_{C1}. That is, the power setting value P_{set} of

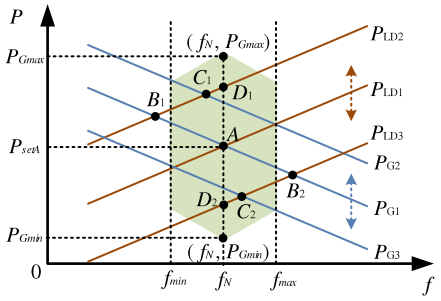


Fig. 8. Secondary frequency regulation diagram.

secondary frequency regulation needs to be set according to the generation margin of the current unit, as shown in Eq. (13). If $P_{\mathrm{LD2}}(f_{\mathrm{N}})$ is less than $\sum_{i=1}^{m} P_{\mathrm{Gmaxi}}$, the generation margin of diesel generator vehicles is not enough to restore the system frequency to the rated value (point D_1), but it can be restored to the normal range, with $f_{\min} < f_{\mathrm{C1}} < f_{\mathrm{N}}$. If $P_{\mathrm{LD2}}(f_{\mathrm{N}})$ is greater than $\sum_{i=1}^{m} P_{\mathrm{Gmaxi}}$, the system frequency can be restored to the rated value f_{N}. The

green shaded part in the figure is the movable range of unit frequency. We have

$$\sum_{i=1}^{m}\left(P_{\text{seti}}^{C}\right)=\begin{cases}\sum_{i=1}^{m}\left(P_{\text{Gmaxi}}\right),\ P_{\text{LD2}}(f_{\text{N}})>\sum_{i=1}^{m}\left(P_{\text{Gmaxi}}\right)\\ P_{\text{LD2}}(f_{\text{N}}),\ P_{\text{LD2}}(f_{\text{N}})\leq\sum_{i=1}^{m}\left(P_{\text{Gmaxi}}\right)\end{cases}\tag{13}$$

Similarly, when the load decreases, the frequency moves from A to B_2, and then moves to C_2 after secondary frequency regulation. If the unit capacity is sufficient, it can be restored to D_2. The unit power $\sum_{i=1}^{m}P_{seti}^{C}$ is set to

$$\sum_{i=1}^{m}\left(P_{\text{seti}}^{C}\right)=\begin{cases}\sum_{i=1}^{m}\left(P_{\text{Gmini}}\right),\ P_{\text{LD2}}(f_{\text{N}})<\sum_{i=1}^{m}\left(P_{\text{Gmini}}\right)\\ P_{\text{LD2}}(f_{\text{N}}),\ P_{\text{LD2}}(f_{\text{N}})\geq\sum_{i=1}^{m}\left(P_{\text{Gmini}}\right)\end{cases}\tag{14}$$

4. Case study

In order to verify the feasibility of the proposed strategy, an emergency power system with two DFIDG and one synchronous diesel generator is established in Matlab/Simulink based on a typical isolated AC system. As shown in Fig. 9, DG1 and DG2 are DFIDG, and DG3 is SDG. Woodward governor and IEEE type Acia excitation system is adopted for the governor and excitation system of DG3, respectively. T_1∼T_3 are 400V/10 kV step-up transformers and T_4∼T_6 are 10 kV/400V step-down transformers. The capacities of DG1, DG1 and DG3 are 1 MVA, 1 MVA and 2 MVA, respectively. The loads L_1, L_2 and L_3 are 1.35 MW, 1.35 MW and 0.45 MW, respectively, and the rated frequency is 50 Hz. The unit frequency regulation coefficient $P\%/f\%$ is 1/0.05, and the unit inertia coefficient $K_{\text{V}}/P_{\text{N}}$ is 12. Among them, parameters T_{ha}, T_{δ}, T_1, T_2 and T_3 are 0.02 s, 0.023 s, 0.01, 0.02 and 0.2 respectively. H_0 and D_{ω} are 4.6687 kgm^2 and 0.0024.

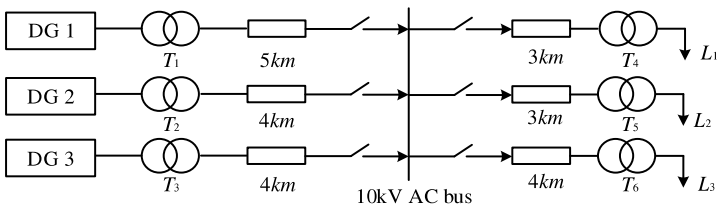


Fig. 9. Structure diagram of a emergency power system.

4.1. Simulation of primary integrated frequency regulation strategy

Load switching test is to verify the ability of emergency microgrid to maintain stable operation in case of large load sudden change. In the initial stage, the output of DG1 and DG2 is stable at 0.6 pu, the output of DG3 is stable at 0.8 pu, and the system frequency is stable at 50 Hz. When the impact load L_3 is connected in the 30 s, the system frequency decreases. Under the same simulation conditions, droop control and primary integrated frequency regulation control are adopted for DFIDG, respectively, and the results are compared.

As shown in Fig. 10(a)∼(d), the frequency changes rapidly when droop control is adopted, and FN drops to 49.48 Hz after 2 s, dΔf/dt changes significantly. When the unit output is at the lowest frequency, the maximum output is 0.8 pu. When the primary integrated frequency control is adopted, FN slowly drops to 49.68 Hz after 4 s, which is close to the stable frequency at the end of primary frequency regulation. The DFIDG responds quickly, provides inertia to the system, slows down system frequency's falling speed, and suppresses frequency oscillation. The DC voltage of a doubly-fed machine can be well maintained by droop control and primary comprehensive frequency control.

As shown in (e) and (f) of Fig. 10, compared with droop control, the electromagnetic torque response of the DFIDG unit with primary comprehensive frequency control is faster, and the overshoot is slight. The mechanical

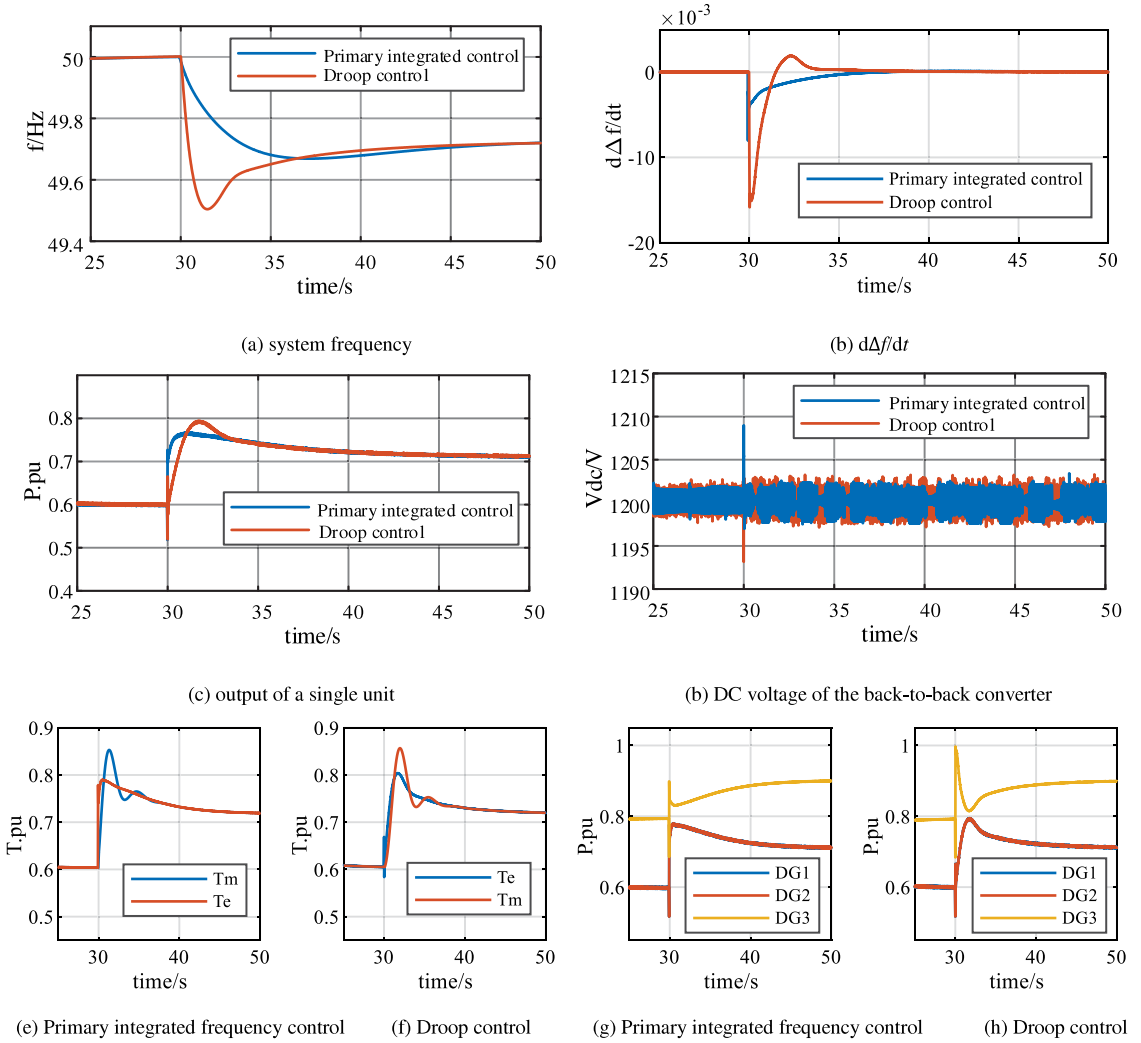


Fig. 10. Comparison diagram of primary integrated frequency control and droop control.

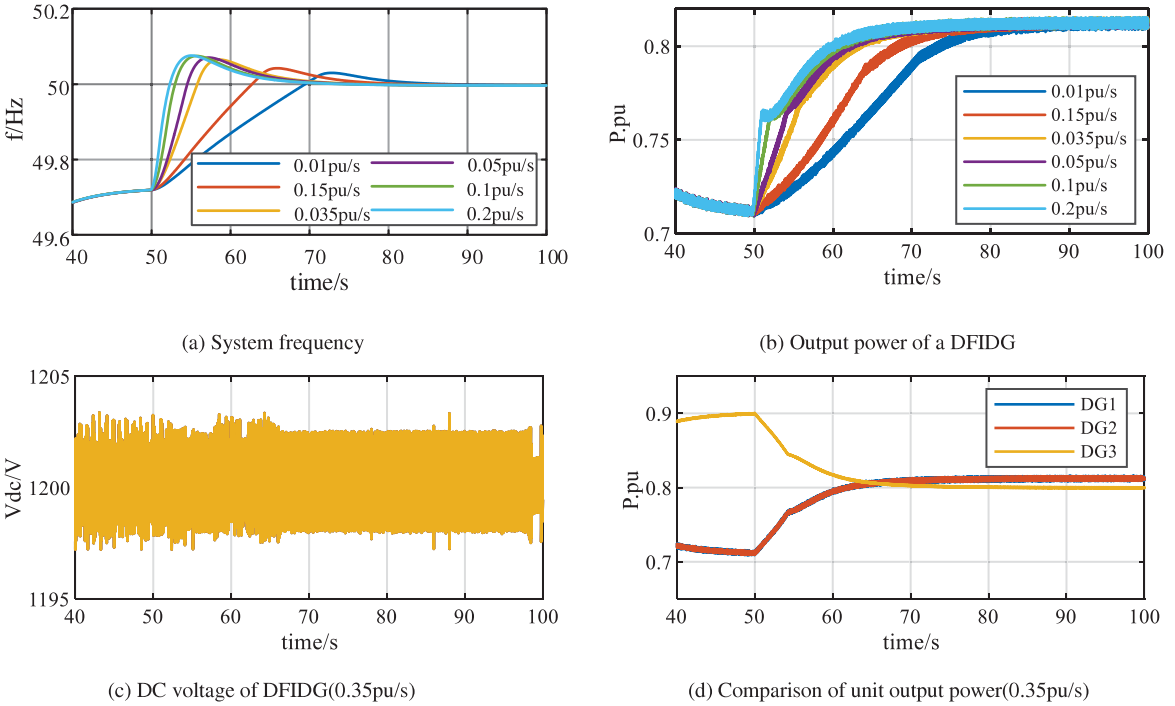
torque lags behind the electromagnetic torque, and its overshoot is also less than the mechanical output torque with droop control. It can be seen that primary integrated frequency control using virtual inertia control can reduce the impact of load switching on the generator and prime mover.

As shown in (g) and (h), when droop control and primary integrated frequency control are adopted, the output of the three DGs does not behave the same in the process of rapid frequency decline. Still, in the steady state, the output of each unit increases by 0.12 pu, and the actually increased output is distributed in proportion to the unit capacity. The equivalent droop control and primary frequency regulation characteristics of the system are verified.

4.2. Simulation of secondary frequency regulation strategy

Since the primary frequency regulation is a non-difference regulation and the system frequency has been stable at about 49.7 Hz, the secondary frequency regulation can also recover the system frequency to the rated value. Based on the maximum output power of DFIDG, system frequency and load power, it can be calculated that the power setting value is 0.8225 pu. In the 50 s, different power adjustment speeds are used for the secondary frequency regulation test.

As shown in Fig. 11(a) and (b), the faster the power adjustment speed is, the quicker the frequency recovery of the system is. But the overshoot is also greater, and they are all lower than 0.1 Hz. The system frequency is restored from the original 49.7 Hz to 50 Hz, and the output of DFIDG is increased by 0.1025 pu, which verifies the effectiveness of the secondary frequency regulation by translation method. It is appropriate to use the speed of 0.035 pu/s for secondary frequency regulation. It can be seen from Fig. 11(c) and (d) that in the process of secondary frequency regulation, the DC voltage of DFIDG is basically unaffected. When the system frequency is stable at 50 Hz, the output of SDG (DG3) not participating in secondary frequency regulation returns to 0.8 pu, and the stable output of DFIDG (DG1 and DG2) is 0.8225 pu, which is consistent with the calculated setting value.



(a) System frequency

(b) Output power of a DFIDG

(c) DC voltage of DFIDG(0.35pu/s)

(d) Comparison of unit output power(0.35pu/s)

Fig. 11. Simulation waveform of secondary frequency regulation strategy.

5. Conclusions

Aiming at the problem of emergency power supply of urban microgrid after disaster, this paper proposes a hierarchical frequency control strategy for DFIDG in an emergency environment without communication. The models of the doubly-fed generator and diesel engine are established, respectively, and the optimal speed power curve is obtained based on the universal characteristic curve of the diesel engine. In order to improve the reliability of the DFIDG emergency power supply, a hierarchical frequency control strategy is proposed, including primary integrated frequency control with droop control and virtual inertia control and secondary frequency regulation control based on translation method. The main conclusions are as follows:

(1) In the process of primary frequency regulation, the DFIDG using virtual inertia control can respond quickly, release rotor kinetic energy to support the frequency, and improve the lowest point of system frequency. The power distribution in the transient state is affected by the virtual inertia of the doubly-fed rotor. However, in steady state, the power variation between DFIDG and other units is still distributed according to capacity.

(2) The secondary frequency regulation control based on translation method can automatically calculate the power setting value of DFIDG online, coordinate DFIDG to distribute dynamic load according to its capacity, and effectively maintain the frequency stability of emergency microgrid.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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